

WASP-43b

R-2002

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_190115 · created 2026-08-07T16:09:32+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458498.8641 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.04
Transit depth (Rp/Rs)^2	3.88 +/- 1.56 [%]
Orbital Inclination (inc)	78.67 +/- 2.35
Airmass coefficient 1 (a1)	1.022 +/- 0.017
Airmass coefficient 2 (a2)	-0.013 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.64 %
Best Comparison Star	#1 - [525, 243]
Optimal Aperture	3.67
Optimal Annulus	8.41
Transit Duration (day)	0.036 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.197 ± 0.04 vs 0.1594 (deviation 0.94σ)
Duration fit vs published	0.036 d vs 0.0512 d (deviation 0.95σ)
Mid-time O–C	2.3 min
Depth SNR	4.9
Residual scatter	1.64 %

Light curve

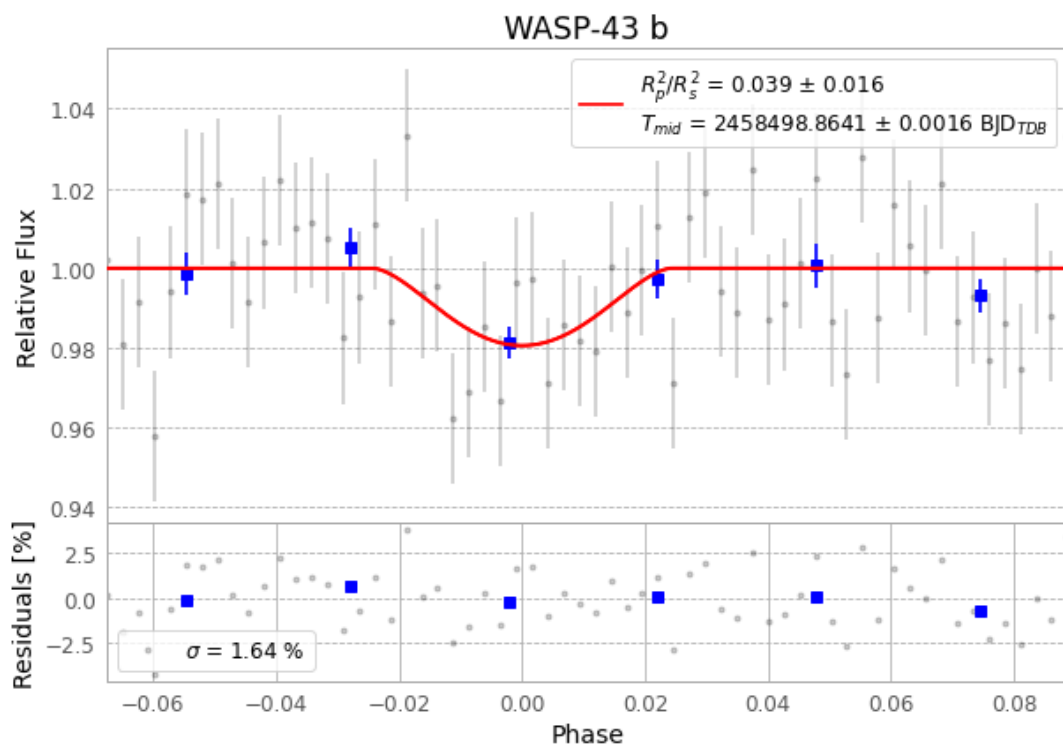


Figure 1 — FinalLightCurve_WASP-43 b_2019-01-15.png (SHA-256 e435441836a5e65a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [369, 214], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5bc537b2d85e837a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), WASP-43190115071212.FITS → WASP-43190115102112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-190115.log (2274965a4b9c...), FinalLightCurve_WASP-43 b_2019-01-15.png (e435441836a5...), FinalLightCurve_WASP-43 b_2019-01-15.csv (eedba3ed9509...)

Compute resources

Wall time 354.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 16:09Z.